

ORDER STATISTICS

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THE DEVIL IS IN THE TAILS

In previous articles, we discussed random variables, probability distributions, and the statistical measures used to describe them. While quantities such as the expected value and variance provide information about the mean and spread of a distribution, they often tell us little about its most extreme outcomes. In many applications, it is precisely these extremes that are of greatest interest. Order statistics are concerned with the values obtained when a sample is arranged from smallest to largest. They provide a foundation for researching measures such as the minimum, maximum, and median. These statistics play an important role in probability and statistics, where understanding the behaviour of extreme observations is often as important as understanding the average outcome.

In risk management, for example, analysts are frequently interested in the largest losses within a portfolio or the distribution of losses beyond a particular threshold. During university, I took a placement year in credit risk, where much of the analysis focused on understanding adverse scenarios and the potential impact of default events. Concepts from order statistics naturally arise in these settings, for example, stress testing metrics, and loss quantiles can all be viewed through the lens of ordered observations.

ORDER STATISTICS

Let $X_1, X_2, \dots, X_n$ be n IID (independent and identically distributed) random variables with cumulative distribution function, CDF, $F_X(x)$ and probability density function, PDF, $f_X(x)$. Arranging the observations from smallest to largest gives the order statistics: $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$, where $X_{(1)}$ is the minimum and $X_{(n)}$ is the maximum observation. Assuming X is continuous, we can derive the distribution function for the minimum $Y_n = X_{(1)} = \min(X_1, X_2, \dots, X_n)$ and for the maximum $Z_n = X_{(n)} = \max(X_1, X_2, \dots, X_n)$ as:

$$P(Y_n \geq x) = P(X_1 \geq x, \dots, X_n \geq x) = (P(X \geq x))^n \quad (1)$$

$$\Rightarrow 1 - F_{Y_n}(x) = (1 - F_X(x))^n \Rightarrow f_{Y_n}(x) = n f_X(x) (1 - F_X(x))^{n-1}$$

$$P(Z_n \leq x) = P(X_1 \leq x, \dots, X_n \leq x) = (P(X \leq x))^n \Rightarrow F_{Z_n} = (F_X(x))^n \Rightarrow f_{Z_n}(x) = n f_X(x) (F_X(x))^{n-1} \quad (2)$$

respectively, by differentiating the CDF to find the PDF. More generally, the k^{th} order statistic $X_{(k)}$ has density:

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F_X(x)]^{k-1} [1 - F_X(x)]^{n-k} f_X(x). \quad (3)$$

WORKED EXAMPLES: EXPONENTIAL AND UNIFORM

Let $X_1, X_2, \dots, X_n$ be n IID random variables with exponential distribution with parameter, λ . What is the CDF, PDF, and expected value of $Y_n = \min(X_1, X_2, \dots, X_n)$? What is the CDF, PDF, and expected value of $Z_n = \max(X_1, X_2, \dots, X_n)$?

Solution: For an exponential distribution, $F_X(x) = 1 - e^{-\lambda x}$, $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$.

Applying $F_X(x)$ and $f_X(x)$ to $Y_n = \min(X_1, X_2, \dots, X_n)$, we have:

$$P(Y_n \geq x) = (P(X \geq x))^n \Rightarrow F_{Y_n}(x) = 1 - (1 - F_X(x))^n = 1 - e^{-n\lambda x}$$

$$\Rightarrow f_{Y_n}(x) = n\lambda e^{-n\lambda x}$$

$$\mathbb{E}[Y_n] = \int_0^\infty x f_{Y_n}(x) dx = \int_0^\infty x n\lambda e^{-n\lambda x} dx = \frac{1}{n\lambda}$$

Applying $F_X(x)$ and $f_X(x)$ to $Z_n = \max(X_1, X_2, \dots, X_n)$, we have:

$$P(Z_n \leq x) = (P(X \leq x))^n \Rightarrow F_{Z_n}(x) = (F_X(x))^n = (1 - e^{-\lambda x})^n$$

$$\Rightarrow f_{Z_n}(x) = n\lambda e^{-\lambda x} (1 - e^{-\lambda x})^{n-1}$$

$$\mathbb{E}[Z_n] = \frac{1}{\lambda} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) = \frac{H_n}{\lambda}$$

where H_n denotes the n^{th} harmonic number.

Let $X_1, X_2, \dots, X_n$ be n IID random variables with uniform distribution between 0 and 1. What is the CDF, PDF, and expected value of $Y_n = \min(X_1, X_2, \dots, X_n)$? What is the CDF, PDF, and expected value of $Z_n = \max(X_1, X_2, \dots, X_n)$?

Solution: For a uniform distribution on $[0, 1]$, $F_X(x) = x$ and $f_X(x) = 1$.

Applying $F_X(x)$ and $f_X(x)$ to $Y_n = \min(X_1, X_2, \dots, X_n)$, we have:

$$\begin{aligned} P(Y_n \geq x) &= (P(X \geq x))^n \Rightarrow F_{Y_n}(x) = 1 - (1 - F_X(x))^n = 1 - (1 - x)^n \\ &\Rightarrow f_{Y_n}(x) = n f_X(x) (1 - F_X(x))^{n-1} = n(1 - x)^{n-1} \end{aligned}$$

$$\mathbb{E}[Y_n] = \int_0^1 n x (1 - x)^{n-1} dx = \int_0^1 n (1 - y) y^{n-1} dy = [y^n]_0^1 - \frac{n}{n+1} [y^{n+1}]_0^1 = \frac{1}{n+1}$$

Applying $F_X(x)$ and $f_X(x)$ to $Z_n = \max(X_1, X_2, \dots, X_n)$, we have:

$$\begin{aligned} P(Z_n \leq x) &= (P(X \leq x))^n \Rightarrow F_{Z_n} = (F_X(x))^n = x^n \\ &\Rightarrow f_{Z_n}(x) = n f_X(x) (F_X(x))^{n-1} = n x^{n-1} \end{aligned}$$

$$\mathbb{E}[Z_n] = \int_0^1 x f_{Z_n}(x) dx = \int_0^1 n x^n dx = \frac{n}{n+1} [x^{n+1}]_0^1 = \frac{n}{n+1}$$

CONCLUSION

Order statistics provides a foundation for analysing the behaviour of ordered observations within a sample. By studying quantities such as the minimum, maximum, and more generally the k^{th} order statistic, we gain insight into the extreme and tail behaviours of random variables. For IID R.V.s, the distributions of these statistics can be derived directly from the underlying distribution, often leading to elegant and useful results. Beyond their theoretical interest, order statistics have numerous practical applications in quantitative finance and risk management, where quantities such as Value-at-Risk (VaR), loss quantiles, and worst-case scenarios are naturally expressed in terms of ordered outcomes.

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